

Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

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Abstract—This study develops a pilot machine learning model for estimating fuel requirements of agricultural machinery in the Philippines. A corpus of 376 cleaned training records was assembled from Agricultural Machinery Testing and Evaluation Center (AMTEC) PDF test reports—including 5 net new records added via Tesseract 5.5.2 OCR over 267 scanned image PDFs (24 candidate extractions, 5 surviving cleaning)—and used to train two models: a Multiple Linear Regression (OLS) baseline and a Random Forest (RF) Regressor. Eight features were used for the RF model—five numeric (`power_kw`, `year`, `abemis_total_count`, `abemis_region_breadth`, `abemis_dominant_region_share`) and three categorical-encoded (`machinery_type`, `machinery_family`, `analysis_subset`). On a held-out test set of 76 records, the RF achieves $R^2 = 0.916$, $MAE = 0.95$ L/h, and $RMSE = 1.53$ L/h, substantially outperforming the OLS baseline. SHAP feature-importance analysis reveals that rated engine power (`power_kw`) dominates at approximately 66.5% of total importance, consistent with the fuel-power proportionality established in the literature. The trained model was applied to score the Agricultural and Biosystems Engineering Management Information System (ABEMIS) national inventory, successfully predicting fuel requirements for 172,265 of 246,741 fuel-relevant records (70%) across all 17 Philippine regions, enabling data-driven national mechanization planning.

Index Terms—Agricultural mechanization, fuel estimation, machine learning, Random Forest, SHAP, LIME, ABEMIS, AMTEC

I. Introduction

Fuel consumption is a critical cost driver in agricultural mechanization and directly affects operational efficiency and sustainability. In the Philippines, fuel requirement estimation is typically based on generalized coefficients or static assumptions that do not account for variability in machine characteristics, field conditions, and operational practices [1].

Traditional engineering-based models provide approximate estimates but are limited in capturing nonlinear relationships among variables such as engine power, implement load, operation type, and field conditions. These models assume deterministic relationships that hold under standardized controlled test conditions; however, actual deployment across the diverse agro-ecological zones of the Philippines introduces variability that fixed coefficients cannot represent [2].

The Agricultural Machinery Testing and Evaluation Center (AMTEC) has accumulated decades of standard-

ized test reports covering a range of machinery types, engine capacities, and operational configurations. Simultaneously, the Bureau of Agricultural and Fisheries Engineering (BAFE) maintains the Agricultural and Biosystems Engineering Management Information System (ABEMIS), a national registry of distributed agricultural machinery. Despite the availability of these datasets, no systematic data-driven fuel estimation framework has been applied to link AMTEC performance data with ABEMIS inventory records.

With the increasing availability of agricultural datasets and the maturation of machine learning tools, it is now feasible to develop predictive models that improve estimation accuracy and support data-driven mechanization planning. This study proposes and evaluates a pilot machine learning model for estimating fuel requirements of selected agricultural machinery operations using structured operational data from AMTEC, enriched with deployment context from ABEMIS.

II. Problem Statement

Current approaches to estimating fuel consumption in agricultural machinery are:

- Based on static or generalized coefficients — fuel requirement tables and engineering formulas apply global constants that are not derived from or validated against local Philippine field conditions.
- Not localized to actual conditions — existing estimates ignore regional variability, machinery age, deployment scale, and crop-stage timing, all of which affect actual fuel demand in practice.
- Unable to capture nonlinear interactions — deterministic formulas treat fuel consumption as a linear function of one or two inputs (typically engine power and area), missing the complex interactions among power, machinery type, operational mode, and deployment context.

As a result, there is a need for a data-driven model that can predict fuel consumption more accurately, adapt to different machine types and operations, and support improved decision-making in mechanized Philippine agriculture.

III. Objectives

General Objective: To develop and evaluate a machine learning model for estimating fuel requirements of selected agricultural machinery operations using AMTEC test data and ABEMIS inventory context.

Specific Objectives:

- Identify key variables affecting fuel consumption in Philippine agricultural machinery;
- Develop a baseline OLS regression model with group-level fallback hierarchy;
- Implement a Random Forest Regressor trained on the same feature set;
- Compare the two models using R^2 , MAE, and RMSE on a held-out test set; and
- Perform feature-importance analysis using SHAP (global) and LIME (local) interpretability methods.

IV. Scope and Delimitation

The study is limited to fuel consumption data extracted from AMTEC PDF test reports and machinery deployment records from ABEMIS. The target variable is fuel consumption in liters per hour (L/h) as reported in standardized AMTEC test conditions. The ABEMIS scoring pipeline applies the trained model to national inventory records to produce region-level fuel requirement estimates.

The cropping calendar from the Regional Agricultural Engineering Division (RAED) was identified as a planned time-dimension predictor representing crop stage and seasonal context. However, the RAED cropping calendar dataset was not yet acquired during the study period and its integration is deferred to future work.

The study excludes real-time sensor acquisition, GPS-based field tracking, full mechanization system simulation, and production-system deployment. Tesseract 5.5.2 OCR was applied to 267 low-text or scanned AMTEC PDFs during the study, producing 24 candidate extractions of which 5 survived cleaning and entered the final corpus (371 $\rightarrow$ 376); the remaining unrecovered reports are predominantly 1990s placeholder filings without machine specifications even after OCR processing.

V. Review of Related Literature

A. Fuel Consumption Theory

Fuel requirement estimation in agricultural mechanization is based on the relationship between engine power, implement load, and operational efficiency. Fuel consumption is generally proportional to the power requirement of the machine during operation, making it a practical indicator of energy demand [1]. Standardized measurement procedures have been developed to characterize this relationship under controlled test conditions, enabling comparison across machine types and power classes.

However, fuel consumption is not determined solely by engine characteristics. It is influenced by implement type,

soil conditions, machine configuration, and operational mode. These relationships indicate that fuel estimation is a multivariate problem, and that single-variable engineering formulas can systematically underperform when applied outside their calibration conditions.

B. Empirical Modeling

Empirical studies show that fuel consumption varies significantly under actual operating conditions. Analysis by PHilMech [3] of commercially available rice combine harvesters demonstrates that fuel consumption is closely related to engine power, field efficiency, and operational parameters including crop variety and moisture content. The Philippine Agricultural Engineering Standard [4] establishes the tractor test procedure that underlies AMTEC reporting, linking rated power and field performance to standardized fuel measurements.

These empirical baselines confirm that while power is the primary determinant, secondary factors modulate actual consumption in ways that motivate a multivariate modeling approach.

C. Real-World Variability

Real-world studies further emphasize that fuel consumption varies across agricultural systems and operating environments. Agaton and Guno [5] report, in the context of irrigation systems, that diesel fuel use is affected by soil type, environmental conditions, crop water requirements, distance to water source, and system efficiency. Although focused on irrigation rather than field machinery, this finding supports the argument that fuel use is context-dependent and that static coefficients cannot generalize across the heterogeneous conditions of Philippine agriculture.

D. Mechanization and Structural Context

Agricultural mechanization increases productivity but also raises energy demand. Mechanization level, often measured in horsepower per hectare, reflects the degree of energy input in agricultural production [6]. Fuel and maintenance costs are major operational barriers faced by farmers, affecting sustainability and mechanization adoption rates [7].

Broader structural constraints—including policy fragmentation, limited financing, and post-harvest infrastructure gaps—also affect resource use and operational efficiency [2], [8]. Historical assessments of the Philippine agricultural machinery industry [9], [10] document the evolution of mechanization from externally funded programs toward domestically driven adoption, setting the institutional context within which AMTEC and BAFE data originate.

E. Research Gap

Existing approaches to fuel estimation rely largely on deterministic formulas, fixed coefficients, and controlled assumptions. These approaches have limited adaptability

across varying field conditions and do not capture interactions among machine, operation, field, and effort-related variables. No prior study has applied machine learning to link AMTEC standardized test data with ABEMIS national inventory records to produce region-level fuel requirement estimates. The present study addresses this gap through a comparative OLS and Random Forest modeling approach with SHAP and LIME interpretability analysis.

VI. Conceptual Framework

Fuel consumption is modeled as a function of four categories of variables:

$$FC = f \left(\begin{array}{l} \text{Machine Characteristics,} \\ \text{Operation Characteristics,} \\ \text{Field/Location Conditions,} \\ \text{Operational Effort} \end{array} \right)$$

The target variable is fuel consumption in L/h as recorded in AMTEC test conditions. The predictor variables in the implemented model are:

- Machine: power_kw (rated engine power in kW), machinery_type (categorical), machinery_family (categorical), analysis_subset (categorical: field vs. stationary/processing)
- Operation/Effort: year (equipment manufacture year, as a proxy for vintage and maintenance-condition cohort)
- Field/Context: abemis_total_count (total national inventory count for the machinery type), abemis_region_breadth (number of regions where the type is deployed), abemis_dominant_region_share (share held by the top deployment region)
- Time (deferred): crop stage and seasonal indicators from the RAED cropping calendar — not yet integrated; reserved for future work

This framework captures machine-level physical determinants as primary predictors, with year encoding equipment-generation effects, and ABEMIS deployment-context features providing secondary structural information about the national inventory landscape. Let the complete feature vector for record i be denoted:

$$\mathbf{x}_i = (x_{\text{power}}, x_{\text{year}}, x_{\text{cnt}}, x_{\text{breadth}}, x_{\text{dom}}, x_{\text{type}}, x_{\text{family}}, x_{\text{sub}})$$

so that the predicted fuel consumption for record i is $\hat{y}_i = \hat{f}(\mathbf{x}_i)$, where $\hat{f}$ is learned from the 300 training records.

VII. Dataset Description

A. AMTEC Test Reports

AMTEC machinery test reports (available at <https://bit.ly/AMTECTestReports2026>) provide the primary source of fuel consumption measurements. Reports cover field machinery, harvesters, dryers, and other mechanization equipment tested under standardized protocols [4]. Text-searchable PDFs were processed using

PyMuPDF to extract fuel consumption (L/h), rated engine power (kW), machinery type, fuel type, field efficiency, operating area, and operating time. Tesseract 5.5.2 OCR was applied to 267 scanned or low-text PDFs, yielding 24 candidate extractions of which 5 survived cleaning and entered the corpus. After unit standardization and outlier filtering, 376 clean records (371 legacy + 5 net new from OCR) were retained as the training and test corpus.

Figure 1 illustrates the strong positive relationship between rated power and fuel consumption across all 376 records, and Figure 2 shows average fuel consumption by machinery family.

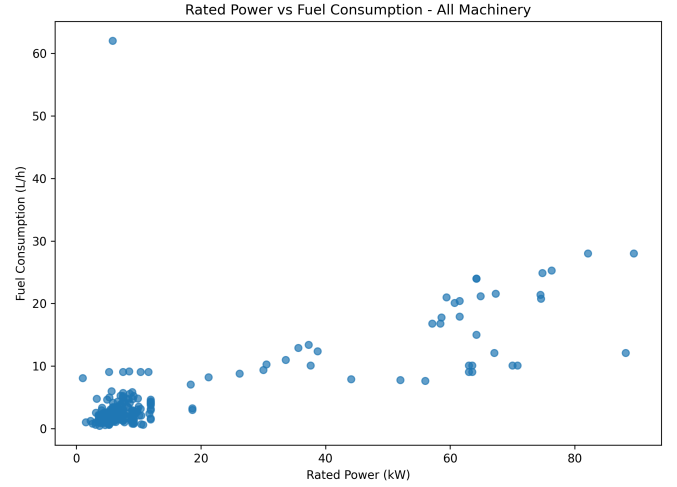


Fig. 1. Rated power (kW) vs. fuel consumption (L/h) for all 376 AMTEC records. A strong positive relationship is visible across all machinery types, supporting power_kw as the dominant predictor.

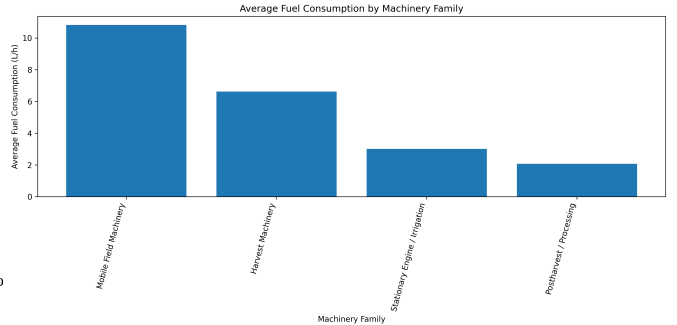


Fig. 2. Average fuel consumption (L/h) by machinery family across the 376 AMTEC records. Harvest machinery and mobile field machinery record the highest average consumption.

B. ABEMIS Inventory

The Agricultural and Biosystems Engineering Management Information System (ABEMIS), accessible at <https://abemis.bafe.gov.ph/user/login>, is maintained by BAFE and records the national distribution of agricultural machinery by region, municipality, barangay, and

recipient type. After filtering for fuel-relevant records (machinery types that consume liquid fuel during operation), 246,741 records were identified across all 17 Philippine regions. Three ABEMIS-derived context features were engineered per machinery type: total inventory count, regional breadth, and dominant-region share.

C. RAED Cropping Calendar (Deferred)

The Regional Agricultural Engineering Division cropping calendar was identified as the source of time-dimension predictors (crop stage, planting/harvesting window) that would link fuel demand to the agricultural season. The RAED dataset was not acquired during the study period. The data gathering instrument (Data Gathering Form for Cropping Calendar, AO 07-22-2022) is on file but the dataset itself is pending. Integration is deferred to future work.

D. Feature Summary

The Random Forest model uses 8 features:

- Numeric (5): power_kw, year, abemis_total_count, abemis_region_breadth, abemis_dominant_region_share
- Categorical-encoded (3): machinery_type, machinery_family, analysis_subset
- Target: fuel consumption (L/h)

VIII. Methodology

A. Pipeline Overview

The modeling pipeline proceeds as follows:

- 1) PDF Extraction — PyMuPDF (fitz) is used to extract structured text from text-searchable AMTEC PDFs. Scanned image PDFs are flagged for deferred OCR processing.
- 2) Unit Standardization — Horsepower values are converted to kW; area units normalized to hectares; time to hours; fuel to liters.
- 3) Outlier Filtering — Records with implausible fuel consumption values (e.g., negative, zero, or extreme outliers by machinery type) are removed using interquartile-range-based filtering at the machinery-type level.
- 4) Feature Engineering — ABEMIS context features are joined to AMTEC records by machinery_type key. Categorical variables are label-encoded for RF input.
- 5) Train/Test Split — An 80/20 random split (376 total records; 300 training, 76 test) is applied with random_state = 42.
- 6) Model Training — OLS baseline and RF Regressor are fitted on the training set.
- 7) Interpretability — SHAP and LIME analyses are performed on the RF model.
- 8) ABEMIS Scoring — The trained RF model scores all 246,741 ABEMIS fuel-relevant records to produce region-level fuel estimates.

B. Baseline OLS Model

A Multiple Linear Regression baseline is implemented with power_kw as the primary predictor. Three functional forms are evaluated for each scope:

Linear:

$$\hat{y} = \beta_0 + \beta_1 \cdot x_{\text{power}}$$

Quadratic:

$$\hat{y} = \beta_0 + \beta_1 \cdot x_{\text{power}} + \beta_2 \cdot x_{\text{power}}^2$$

Log-response:

$$\ln(\hat{y}) = \beta_0 + \beta_1 \cdot x_{\text{power}} \implies \hat{y} = \exp(\beta_0 + \beta_1 \cdot x_{\text{power}})$$

Parameters β are estimated by minimizing the residual sum of squares (OLS):

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

To handle machinery-type heterogeneity, a group-level fallback hierarchy is applied: the model first uses a machinery_type-level fit; if insufficient data exist for a given type ($n < 5$), it falls back to the machinery_family-level fit; and if that also fails, it falls back to the global (all-machinery) model. Only models passing the adjusted- R^2 filter threshold (adjusted $R^2 \geq 0.50$) are retained in the prediction hierarchy.

C. Random Forest Regressor

A Random Forest Regressor is trained using scikit-learn with hyperparameters: n_estimators=300, max_depth=None, min_samples_split=5, min_samples_leaf=2, random_state=42, n_jobs=-1. All 8 features are used. Label encoders are fitted per categorical feature on the training set and applied to both test and ABEMIS scoring sets. The model is evaluated on the 76-record holdout set and validated with 5-fold cross-validation (shuffle=True, random_state=42).

The ensemble prediction for record i is the average of $B = 300$ regression trees:

$$\hat{y}_i = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}_i)$$

where $T_b(\mathbf{x}_i)$ denotes the prediction of the b -th decision tree, each grown on a bootstrap sample of the training data with a random subset of features considered at each split.

D. Interpretability

SHAP TreeExplainer is applied to the trained RF model. SHAP decomposes the prediction for each record i into a sum of feature contributions:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^p \phi_{ij}$$

where ϕ_0 is the baseline (mean prediction over the training set), $p = 8$ is the number of features, and ϕ_{ij} is the SHAP

value for feature j and record i . Global feature importance is measured as:

$$I_j = \frac{1}{n} \sum_{i=1}^n |\phi_{ij}|$$

Global feature importance (mean absolute SHAP value per feature) and per-sample SHAP contributions are computed. A beeswarm plot visualizes the distribution of SHAP values across all test samples. LIME tabular explainer generates local explanations for 10 selected example records, identifying which features most influenced each individual prediction. Results are exported to SHAP_Analysis.xlsx and LIME_Explanations.xlsx.

E. Evaluation Metrics

Model performance is evaluated using:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

where y_i is the actual fuel consumption, $\hat{y}_i$ is the model prediction, $\bar{y}$ is the sample mean, and $n = 76$ for the holdout set.

IX. Results

A. Model Comparison

Table I presents the holdout evaluation metrics for the OLS and Random Forest models. The Random Forest substantially outperforms all OLS variants across all three metrics.

TABLE I
OLS vs. Random Forest holdout metrics ($n = 76$)

Model	R^2	MAE	RMSE
OLS Global Linear	0.704	1.88	2.88
OLS Hierarchical FB	0.703	1.88	2.88
Random Forest	0.916	0.95	1.53

L/h = liters per hour

Note: Hierarchical OLS uses a per-machinery-type fallback chain (type $\rightarrow$ family $\rightarrow$ global). Random Forest 5-fold cross-validation mean $\pm$ std: $R^2 = 0.894 \pm 0.023$, $\text{MAE} = 1.11 \pm 0.16$ L/h, $\text{RMSE} = 1.78 \pm 0.36$ L/h. Full results are in OLS_vs_RF_Comparison.xlsx.

B. SHAP Feature Importance

SHAP analysis (Figures 4–5 and Figure 3 in the main text) reveals that power_kw dominates feature importance at approximately 66.5% of total mean absolute SHAP value. The year feature is second at approximately 12.3%, capturing vintage-related differences in fuel efficiency across equipment cohorts. Categorical features (machinery_type, machinery_family, analysis_subset) account for approximately 12.1% combined. The three ABEMIS context features (abemis_total_count, abemis_dominant_region_share, abemis_region_breadth) contribute approximately 9.1% combined, with abemis_region_breadth negligible ($< 0.2\%$) but abemis_total_count and abemis_dominant_region_share contributing a consistent 4.7% and 4.2% respectively. Full values are reported in Table II.

TABLE II
SHAP global feature importance (mean |SHAP|) for the Random Forest model

Feature	Mean SHAP	%
power_kw	3.256	66.5%
year	0.600	12.3%
machinery_type	0.253	5.2%
abemis_total_count	0.232	4.7%
abemis_dominant_region_share	0.207	4.2%
analysis_subset	0.204	4.2%
machinery_family	0.137	2.8%
abemis_region_breadth	0.005	0.1%

The dominance of power_kw is consistent with the physical fuel–power proportionality established by Domsch et al. [1]: higher engine power ratings require more fuel to sustain operation regardless of machinery type. The secondary importance of year reflects the improvement in fuel efficiency across equipment generations and is physically interpretable.

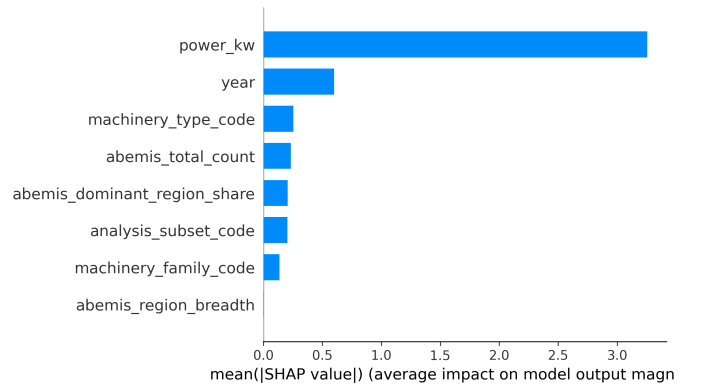


Fig. 3. SHAP global feature importance (mean |SHAP|) for the Random Forest model. power_kw accounts for approximately 66.5% of total importance; year is second at 12.3%. Appendix A shows the full beeswarm plot.

C. LIME Local Explanations

LIME tabular explainer was applied to 10 example records (Appendix C). Across all 10 instances, `power_kw` appeared as the top local contributor in 9 out of 10 cases. Machinery type and `analysis_subset` provided secondary adjustments, particularly for records involving dryers and stationary processing equipment. The LIME results are consistent with the global SHAP picture, providing per-record interpretability for model validation.

D. ABEMIS National Scoring

The trained RF model was applied to all 246,741 fuel-relevant ABEMIS records to produce regional fuel requirement estimates:

- Successfully scored: 172,265 records (70%) across all 17 Philippine regions
- Excluded — missing Rated Power: 62,808 records (~25%) where the ABEMIS machinery entry did not carry a parseable rated power value
- Excluded — unmappable machinery type: 11,668 records (~5%) where the ABEMIS machinery type could not be mapped to an AMTEC-covered type

Regional fuel estimates and dominant machinery families are reported in `ABEMIS_Regional_Fuel_Estimates.xlsx`. Mobile field machinery (tractors, tillers) and harvest machinery (combine harvesters, reapers) account for the largest aggregate regional fuel estimates, consistent with their dominant share of the ABEMIS inventory.

X. Discussion

A. Primacy of Engine Power

The dominant contribution of `power_kw` (~66.5% SHAP importance) confirms the fundamental physical relationship between engine capacity and fuel demand [1]. This result holds across all SHAP analysis levels (global bar chart, beeswarm distribution, and individual LIME explanations), providing high confidence in the model’s physical coherence. The secondary importance of `year` (~12.3%) is also physically interpretable: newer equipment generations incorporate improved combustion technology and tighter manufacturing tolerances, resulting in systematically lower fuel intensity at equivalent rated power. The quadratic OLS enrichment of `power_kw` also improved OLS performance over the linear baseline, further validating the nonlinear power–fuel relationship.

B. ABEMIS Context Features

The three ABEMIS deployment-context features contribute approximately 9.1% of total SHAP importance combined (`abemis_total_count`: 4.7%, `abemis_dominant_region_share`: 4.2%, `abemis_region_breadth`: < 0.2%). This suggests that knowledge of how widely and densely a machinery type is deployed across the country provides marginal but real predictive value, possibly capturing the fact that

machinery types with higher national inventory counts tend to be better-maintained or operated at higher utilization rates. This contribution is not large enough to substitute for direct power measurement but adds value in scoring scenarios where power alone is insufficient.

C. Limitations

Several limitations constrain the scope and generalizability of the current results:

- Corpus size: Tesseract 5.5.2 OCR was applied to 267 previously low-text or unreadable AMTEC reports during this study, yielding 24 candidate extractions of which 5 survived cleaning (corpus expanded from 371 legacy to 376 final, i.e. 371 legacy + 5 net new from OCR). The remaining unrecovered reports are predominantly 1990s placeholder filings without machine specifications even after OCR. Acquisition of additional source PDFs (notably the previously inaccessible 3,065-PDF collaborator set) would further enlarge the corpus and improve coverage of underrepresented machinery types.
- RAED cropping calendar unavailable: The time-dimension predictor (crop stage, planting/harvest window) could not be incorporated because the RAED dataset was not yet acquired. This limits the model’s ability to account for seasonal load variations.
- Missing Rated Power in ABEMIS: Approximately 25% of ABEMIS records (62,808 rows) carry no parseable Rated Power value, making them unscorable by the current model. Imputation or supplementary data linkage would be required to recover these rows.
- Machinery-type mapping gaps: Certain ABEMIS machinery types (e.g., corn mill) were approximated using the nearest AMTEC-tested proxy (e.g., rice mill). These proxy mappings introduce uncertainty and account for the 11,668 currently-unmapped records.

XI. Conclusion

This pilot study demonstrates that a Random Forest Regressor trained on 376 AMTEC test records achieves strong holdout performance ($R^2 = 0.916$, $MAE = 0.95$ L/h, $RMSE = 1.53$ L/h), substantially outperforming the deterministic OLS baseline across all three evaluation metrics. The trained model successfully scores 172,265 ABEMIS deployment records across all 17 Philippine regions, producing the first data-driven, region-level fuel requirement estimates for national agricultural machinery inventory. SHAP and LIME interpretability analyses confirm that the model’s behavior is physically coherent, with engine power as the dominant driver and deployment-context features providing consistent secondary contributions. The results demonstrate the feasibility of linking AMTEC test data to ABEMIS inventory for operational planning, and provide a principled foundation for expanding the approach as additional data sources—including scanned

AMTEC reports and the RAED cropping calendar—become available.

XII. Recommendations and Future Work

The most consequential next steps move this pilot from a single trained model into a planning instrument. Three directions are prioritized; data-quality remediation is treated as supporting work that enables them.

A. Priority Directions

- National fuel forecasting. Operationalize the trained RF model as a recurring, region-level fuel demand estimator for BAFE planning workflows. Concretely, this means scoring updated ABEMIS snapshots on a routine schedule, publishing per-region and per-machinery-family demand summaries, and integrating the outputs into fuel-assistance allocation and mechanization budgeting cycles. The 70% national coverage already achieved is sufficient to support an initial planning release.
- Seasonal planning via the RAED cropping calendar. Acquire and ingest the Regional Agricultural Engineering Division cropping calendar so that crop-stage timing enters the model as time-dimension predictors (`is_in_season`, `crop_stage_intensity`). This converts the current static, per-record fuel estimate into a season-resolved regional demand forecast — the single most important missing predictor and the largest available gain in planning relevance.
- Real-world validation via in-field telemetry. Run a pilot deployment of GPS- and flow-sensor-instrumented machinery across selected regions and compare measured fuel use against AMTEC-trained predictions. This closes the lab-to-field loop that the AMTEC corpus alone cannot, calibrates regional adjustment factors, and produces the empirical evidence base required for institutional adoption.

B. Supporting Data-Quality Work

The following are enabling activities rather than headline directions. Tesseract 5.5 OCR has already been applied to the 267 scanned AMTEC reports available during this study; acquisition of the additional 3,065-PDF collaborator set and a routine re-extraction would continue to expand corpus coverage. A systematic ABEMIS–AMTEC machinery-type crosswalk would recover the 11,668 currently-unmapped rows, and a rated-power imputation pass (predicting power from machinery type, manufacturer, and model year, or via a machinery registry lookup) would recover the 62,808 ABEMIS records missing parseable power. These mechanical improvements raise national coverage and reduce proxy-mapping uncertainty, but the headline value of the framework comes from the three priority directions above.

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Appendix A SHAP Analysis

SHAP (SHapley Additive Explanations) TreeExplainer was applied to the trained Random Forest model. Figure 4 shows the global feature importance as mean absolute SHAP values. Figure 5 shows the beeswarm plot, illustrating the direction and magnitude of each feature's contribution across all test records.

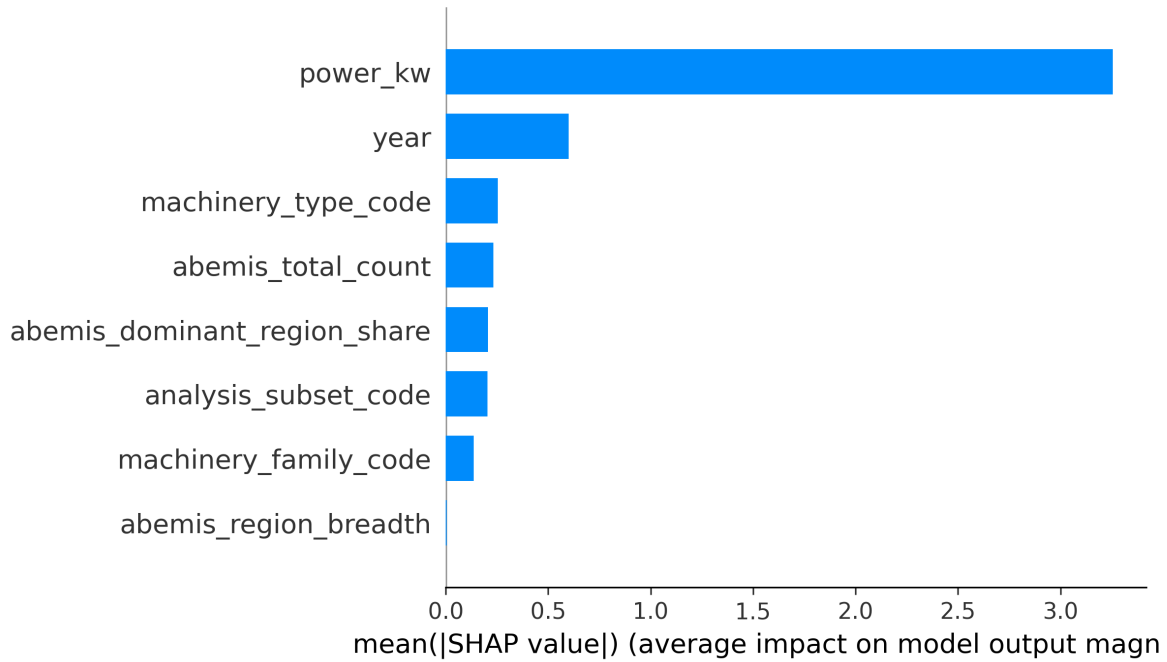


Fig. 4. SHAP summary bar plot — global feature importance (mean $|\text{SHAP}|$) for the Random Forest model. `power_kw` accounts for approximately 66.5% of total importance; `year` is second at 12.3%.

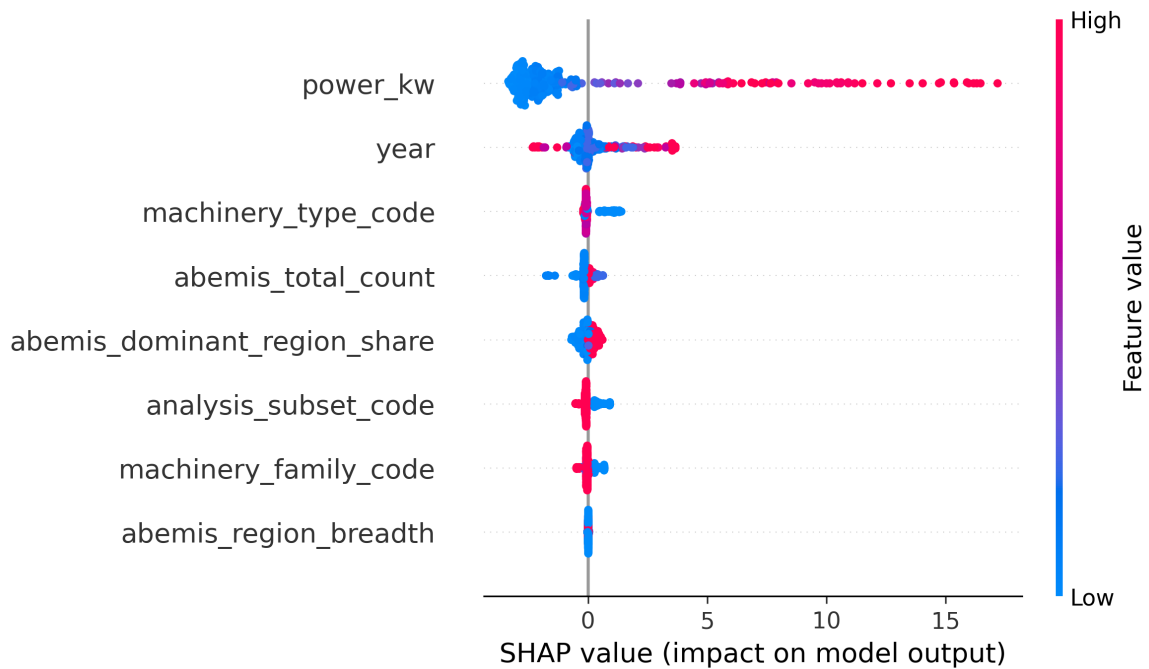


Fig. 5. SHAP beeswarm plot — distribution of SHAP values per feature across all test records. Color indicates feature value magnitude (red = high, blue = low).

Appendix B

OLS Diagnostic Plots

OLS diagnostic plots are shown for six model scopes: two machinery-family scopes and four machinery-type scopes. Each subsection shows four standard diagnostic plots: residuals vs. fitted, Q-Q plot, leverage plot, and studentized residuals plot.

A. Machinery Family: Mobile Field Machinery

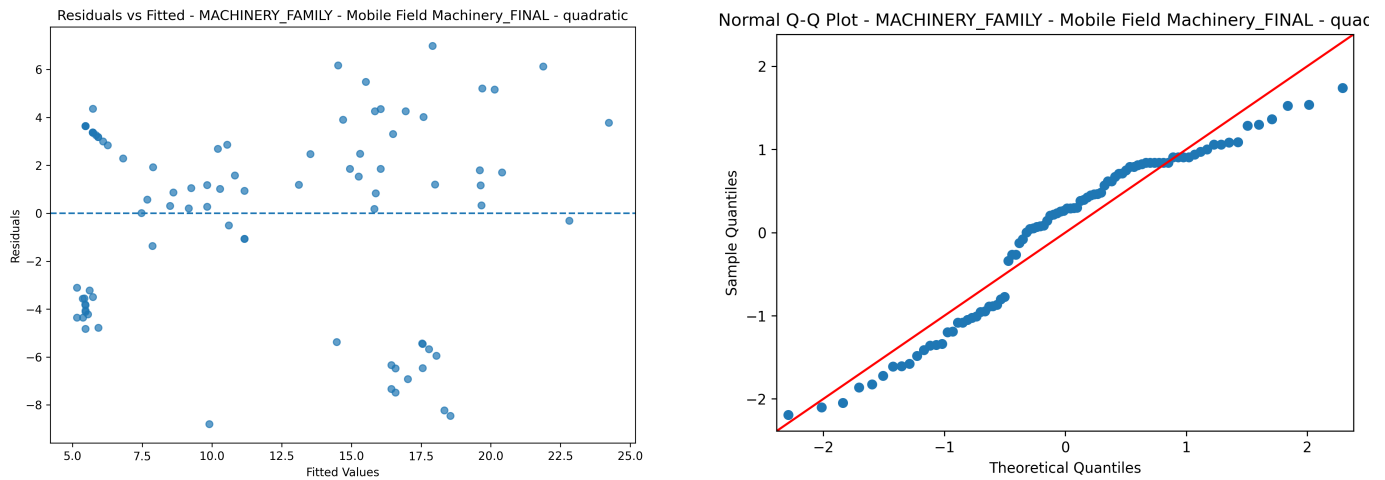


Fig. 6. Mobile Field Machinery — residuals vs. fitted (left) and Q-Q plot (right).

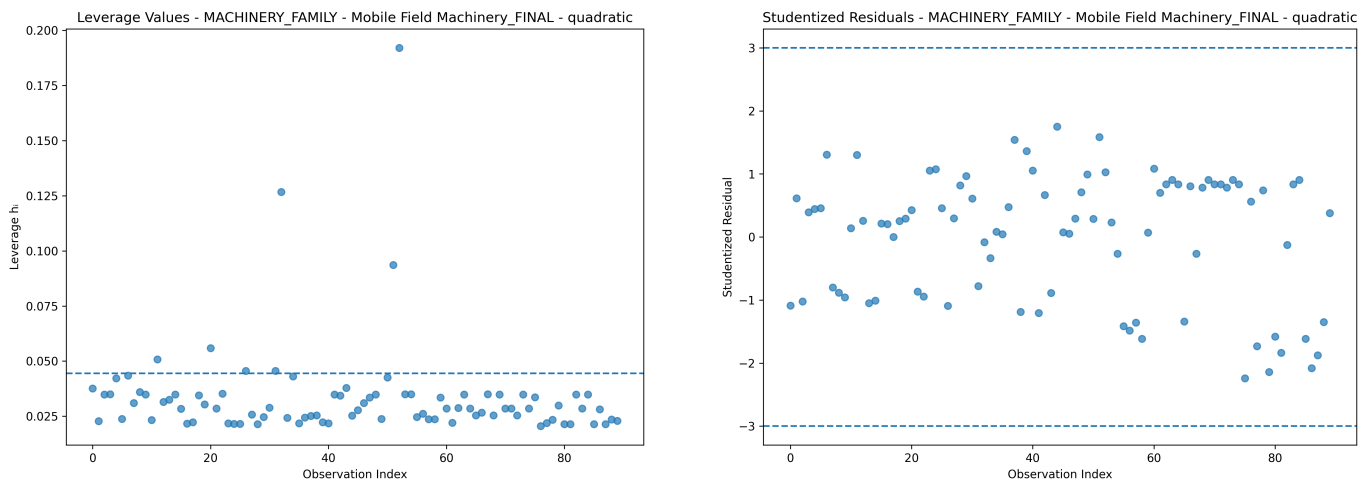


Fig. 7. Mobile Field Machinery — leverage plot (left) and studentized residuals (right).

B. Machinery Family: Harvest Machinery

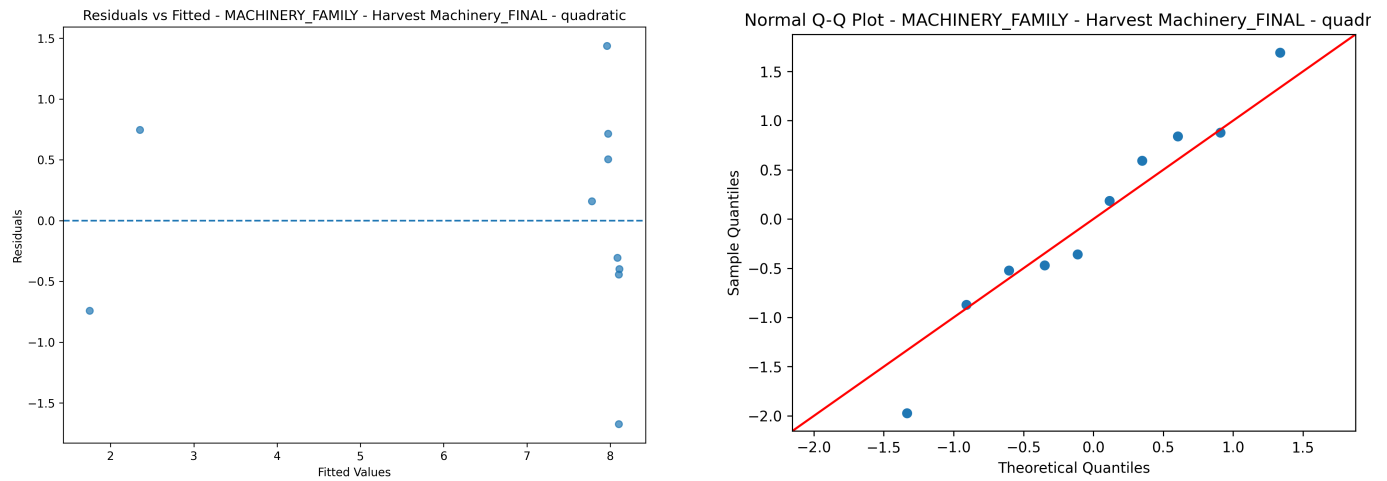


Fig. 8. Harvest Machinery — residuals vs. fitted (left) and Q-Q plot (right).

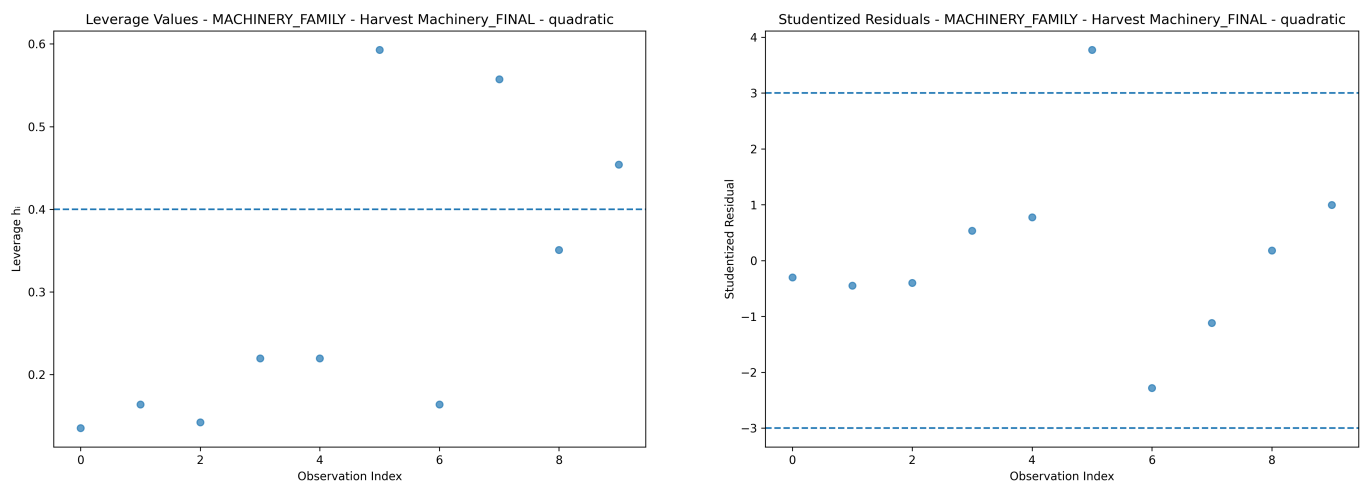


Fig. 9. Harvest Machinery — leverage plot (left) and studentized residuals (right).

C. Machinery Type: Four-Wheel Tractor

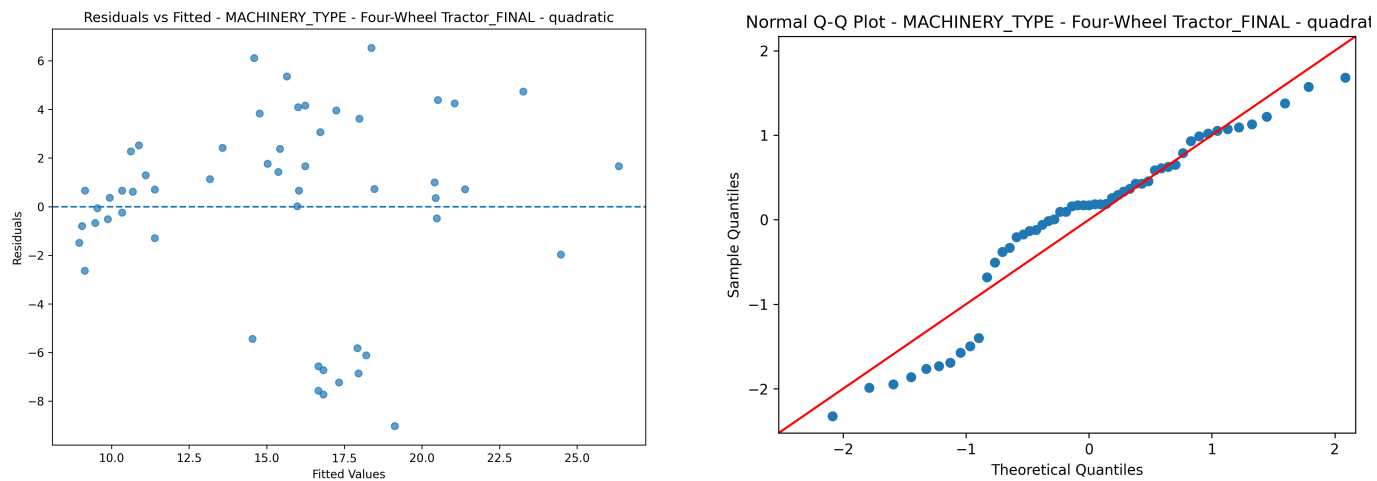


Fig. 10. Four-Wheel Tractor — residuals vs. fitted (left) and Q-Q plot (right).

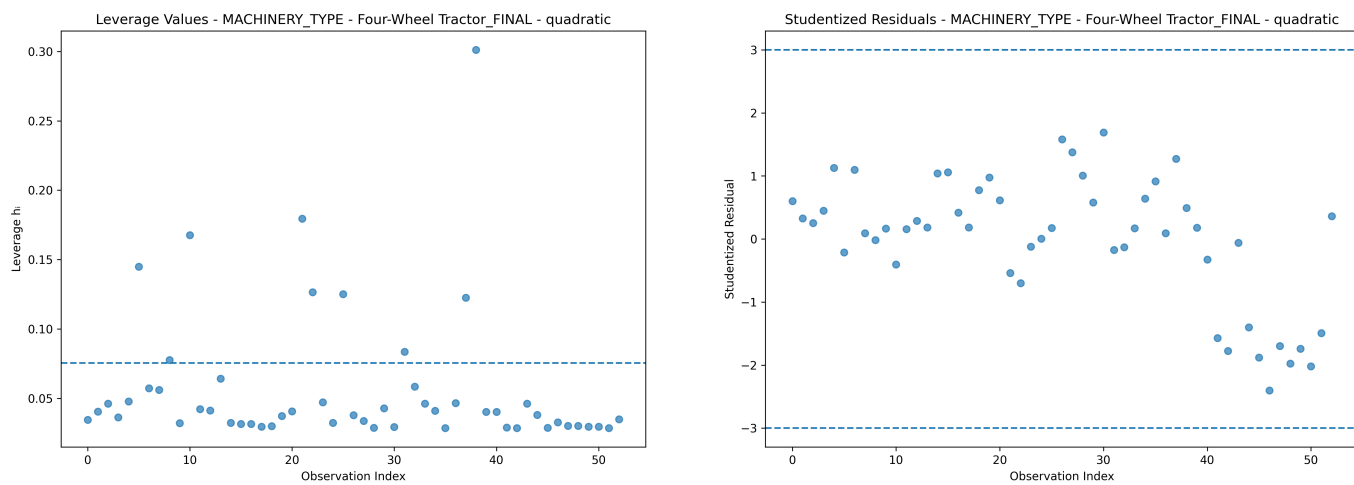


Fig. 11. Four-Wheel Tractor — leverage plot (left) and studentized residuals (right).

D. Machinery Type: Combine Harvester

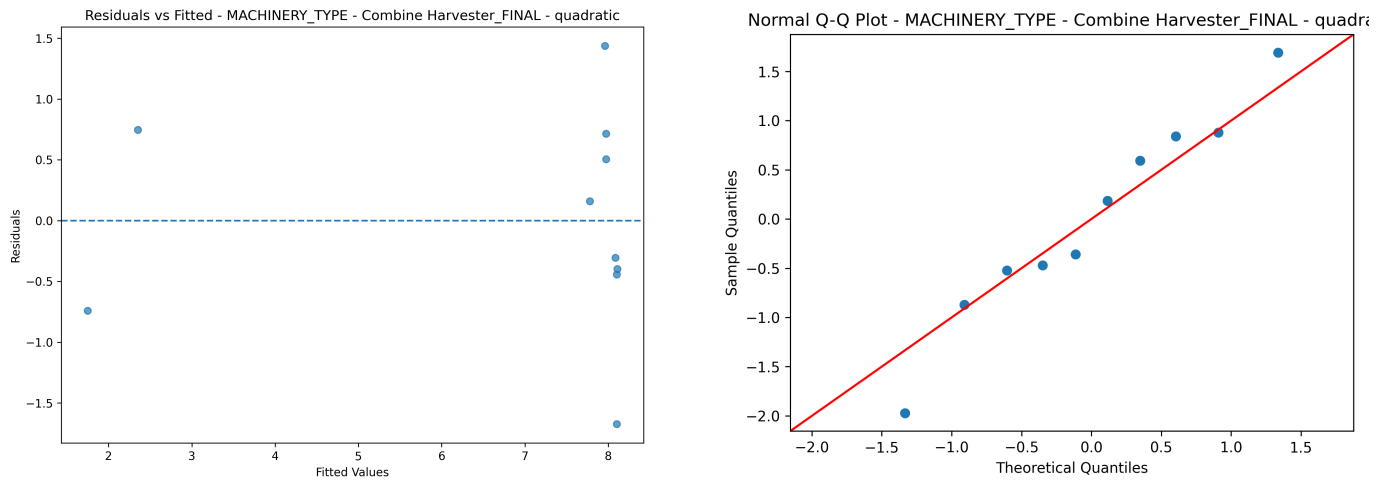


Fig. 12. Combine Harvester — residuals vs. fitted (left) and Q-Q plot (right).

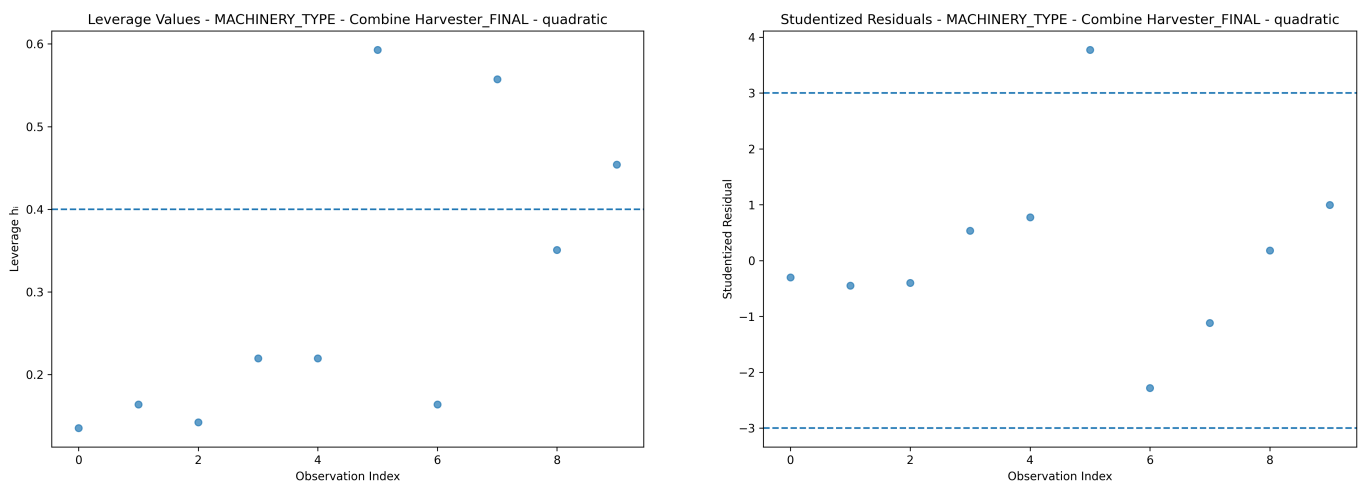


Fig. 13. Combine Harvester — leverage plot (left) and studentized residuals (right).

E. Machinery Type: Mechanical Dryer

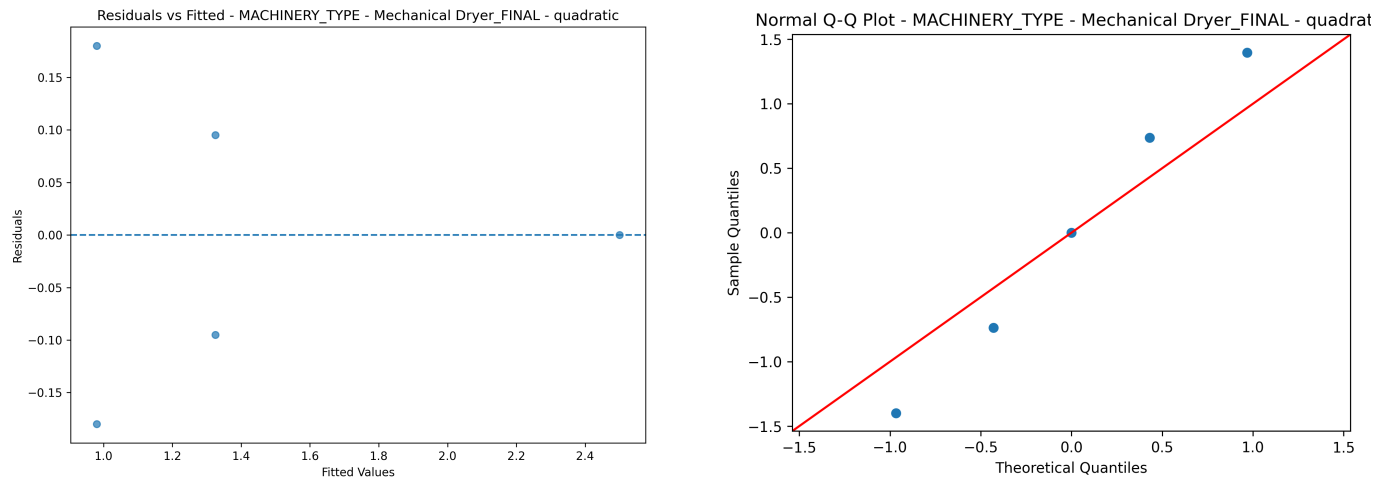


Fig. 14. Mechanical Dryer — residuals vs. fitted (left) and Q-Q plot (right).

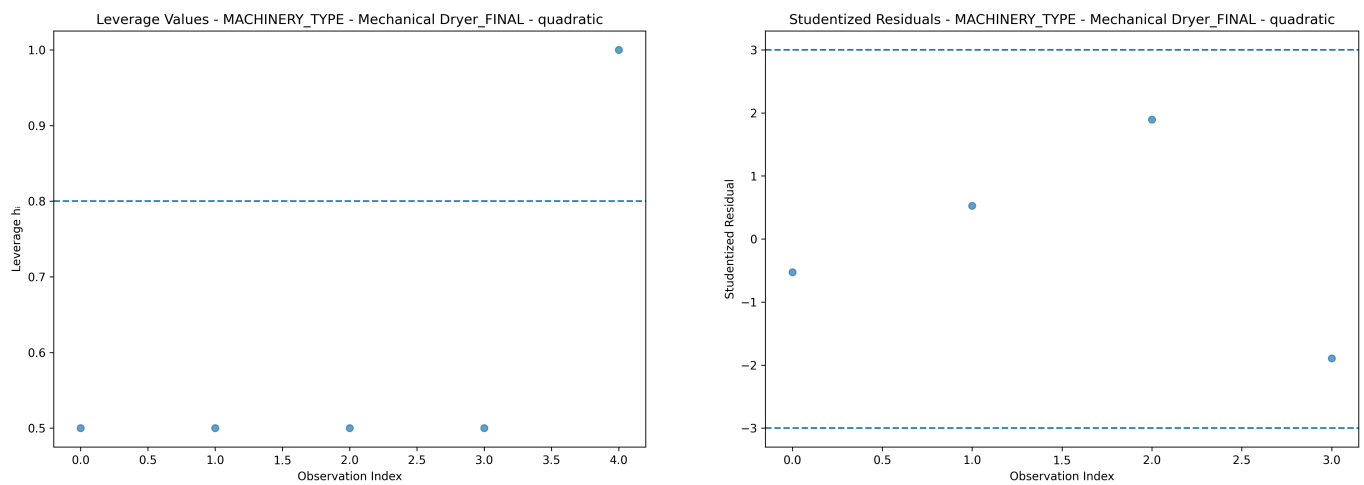


Fig. 15. Mechanical Dryer — leverage plot (left) and studentized residuals (right).

F. Global Model: All Machinery

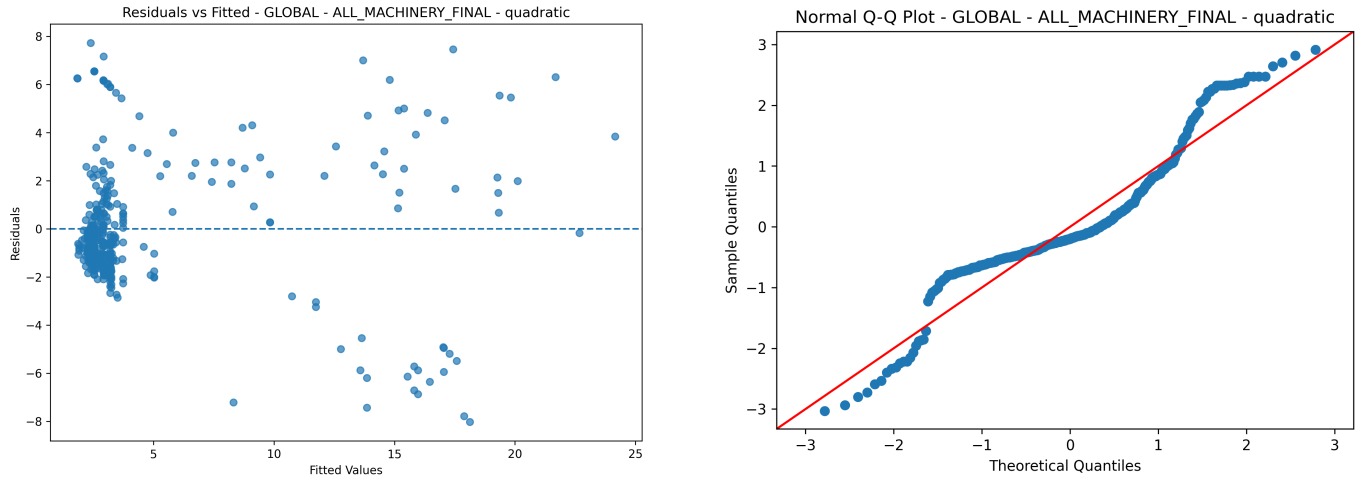


Fig. 16. Global (All Machinery) — residuals vs. fitted (left) and Q-Q plot (right).

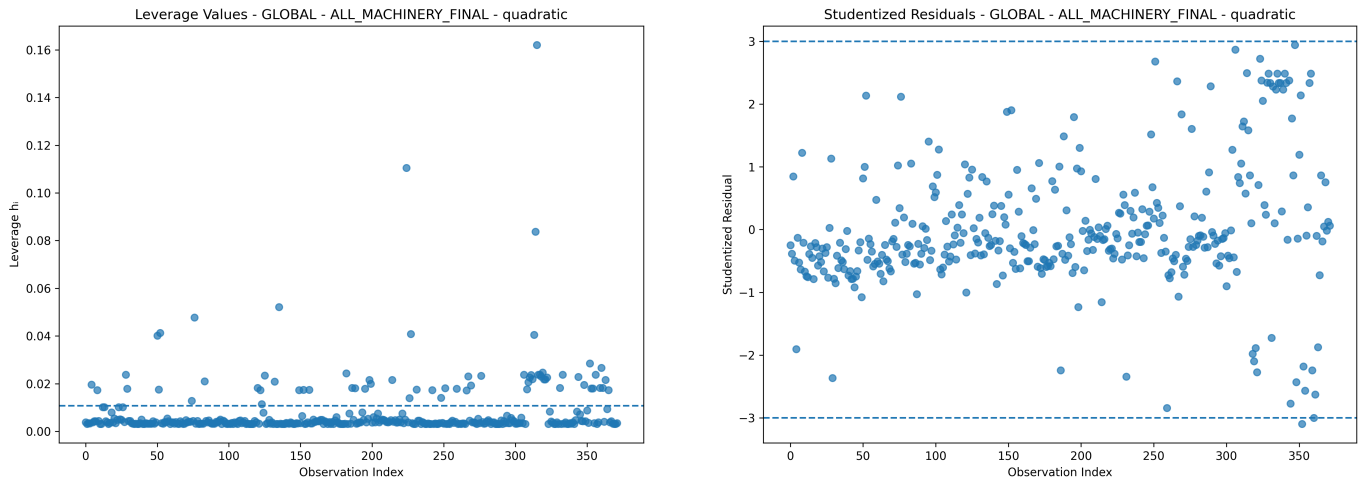


Fig. 17. Global (All Machinery) — leverage plot (left) and studentized residuals (right).

Appendix C LIME Local Explanations

LIME tabular explainer was applied to 10 representative records from the test set. Each figure shows the local feature contributions for one record, with green bars indicating features that increased the predicted fuel consumption and red bars indicating features that decreased it.

LIME - Sample 372 | Small Engine | Actual=5.240 Pred=4.773

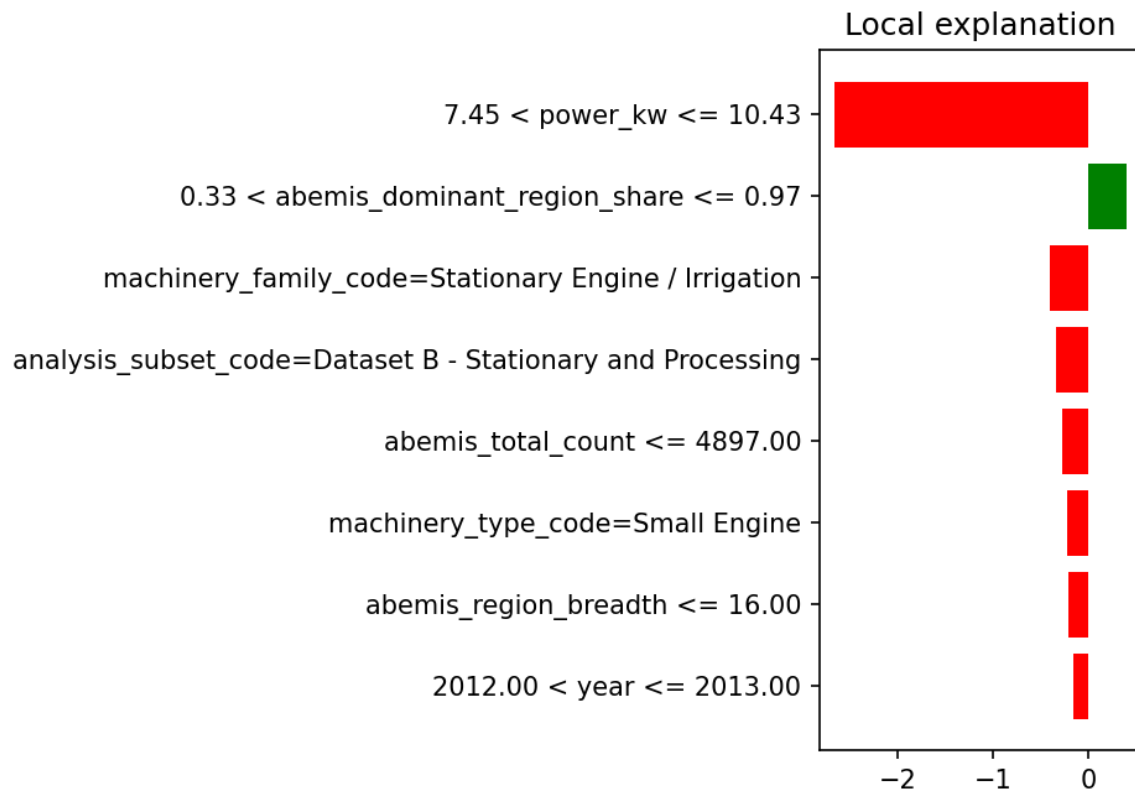


Fig. 18. LIME local explanation — example record 0.

LIME - Sample 284 | Small Engine | Actual=3.560 Pred=3.670

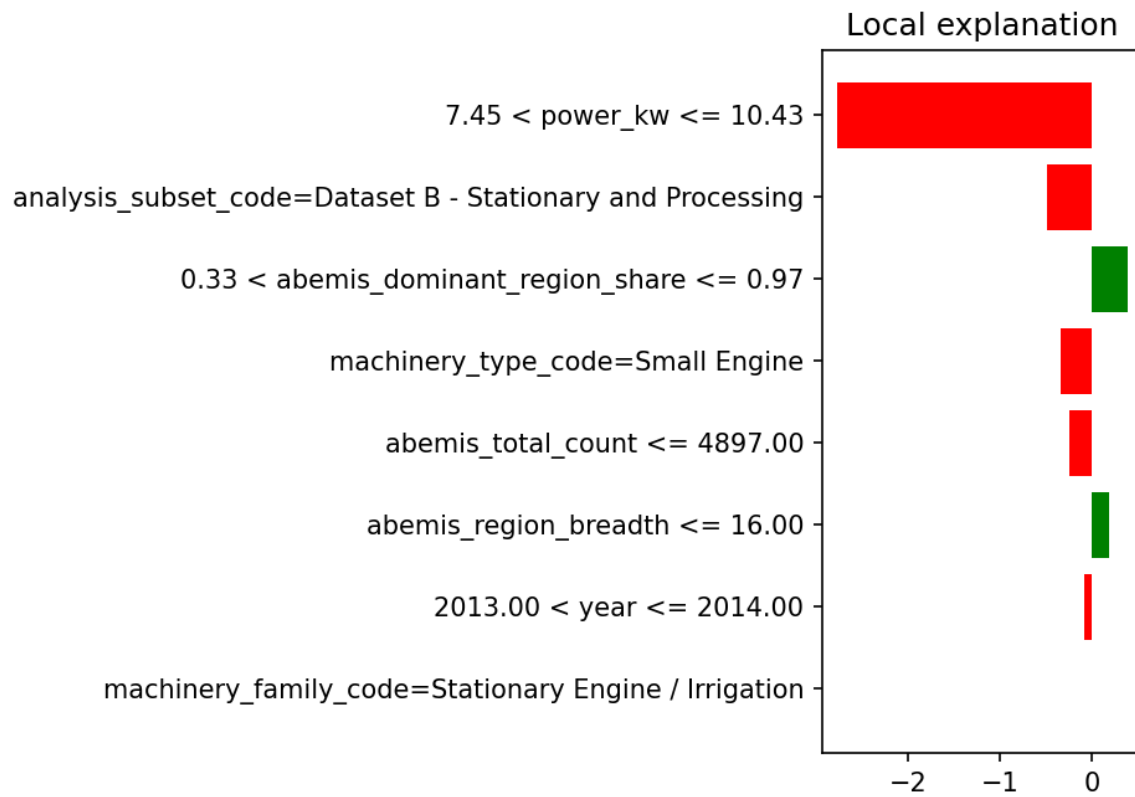


Fig. 19. LIME local explanation — example record 1.

LIME - Sample 32 | Thresher | Actual=1.570 Pred=1.601

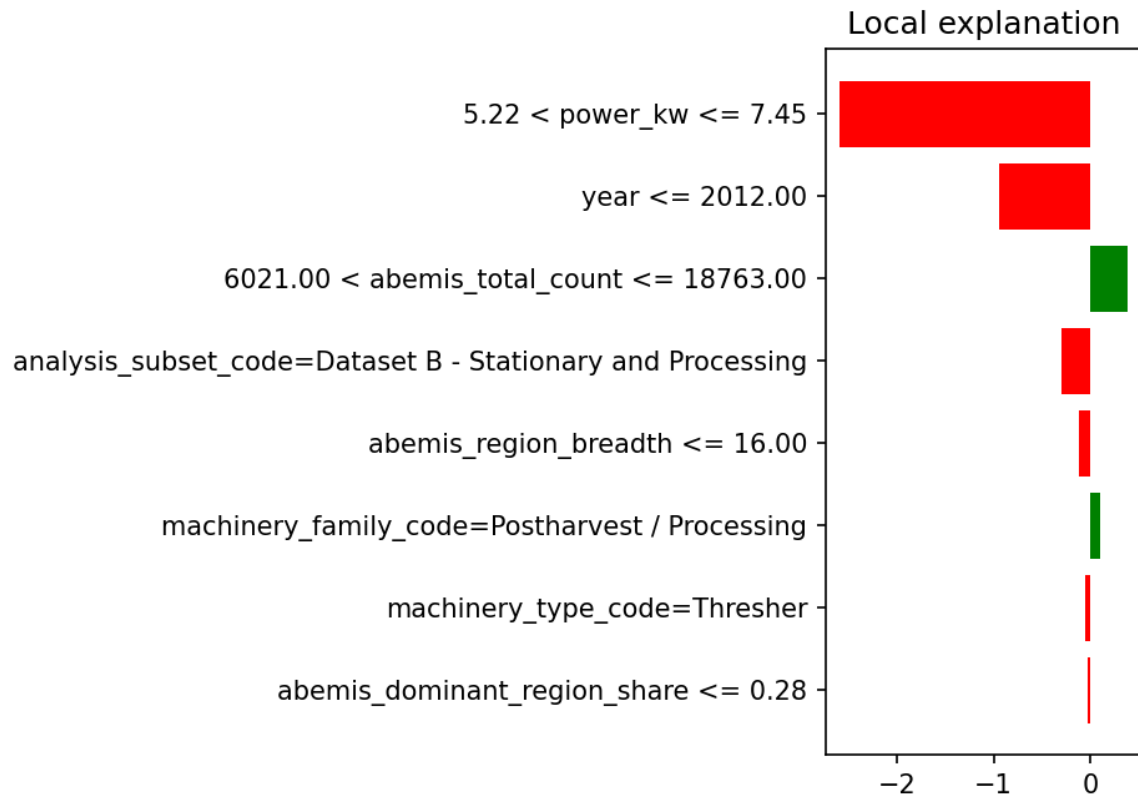


Fig. 20. LIME local explanation — example record 2.

LIME - Sample 241 | Water Pump | Actual=2.330 Pred=2.115

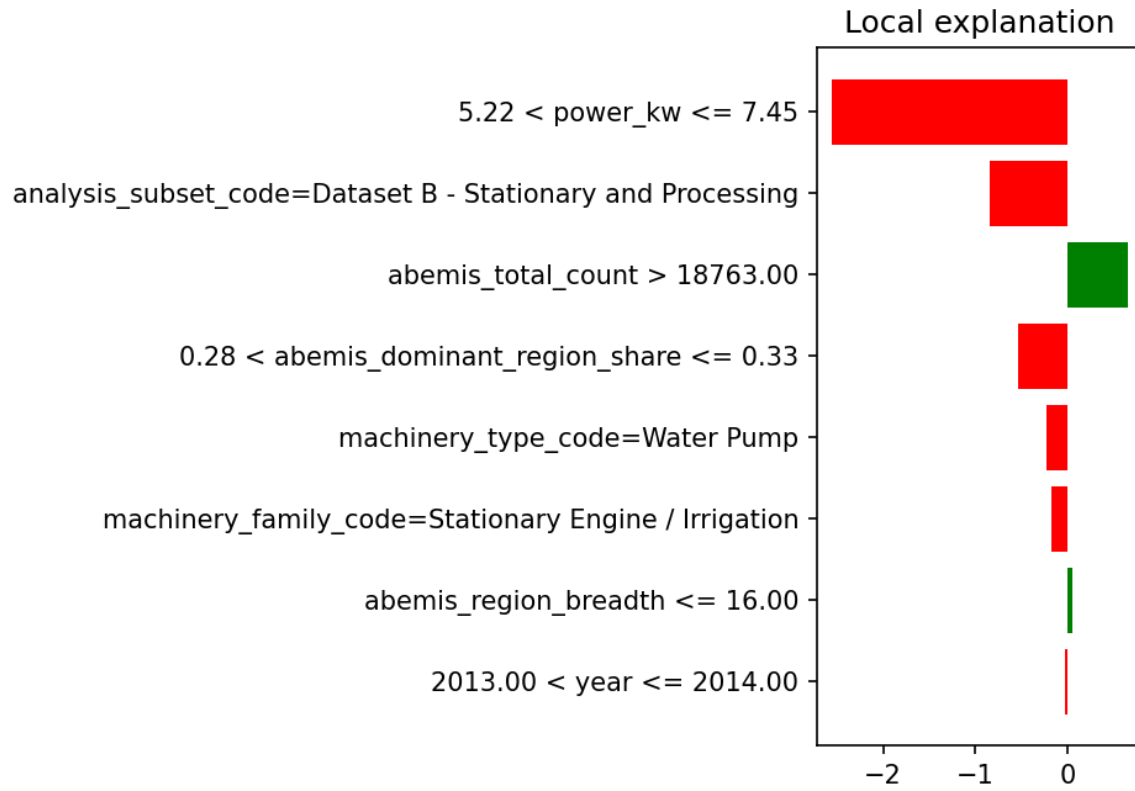


Fig. 21. LIME local explanation — example record 3.

LIME - Sample 162 | Small Engine | Actual=1.770 Pred=1.972

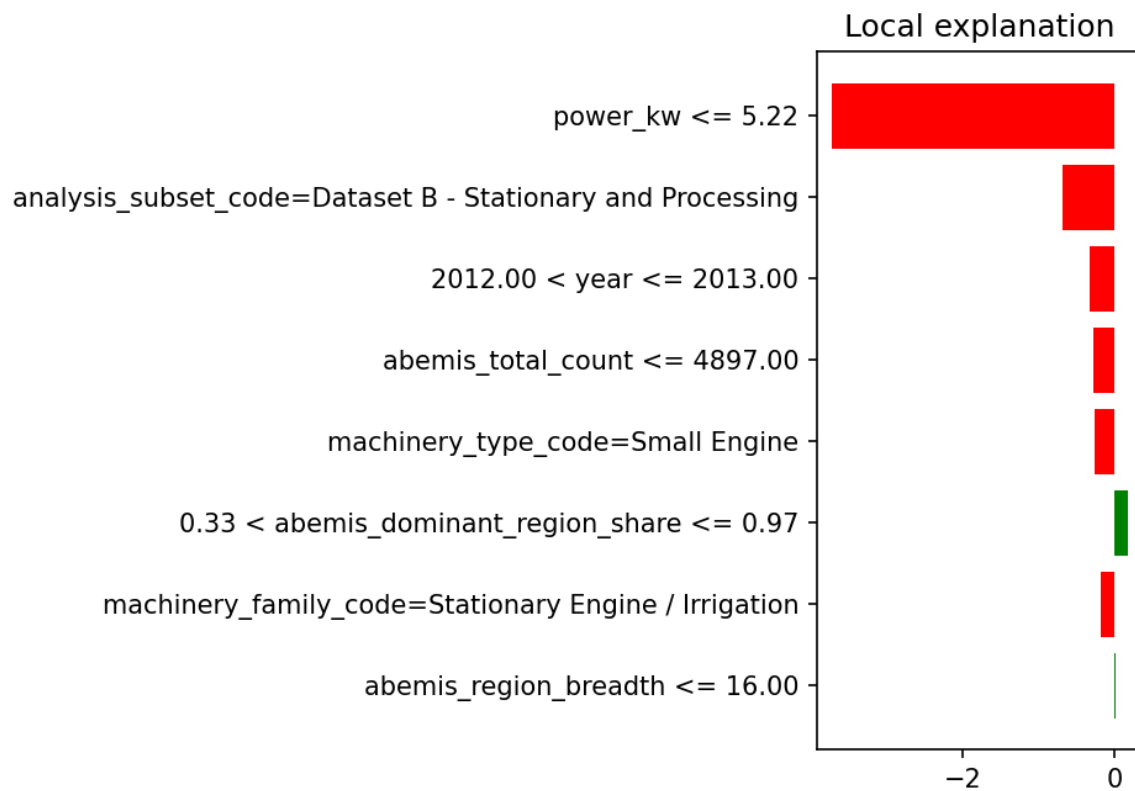


Fig. 22. LIME local explanation — example record 4.

LIME - Sample 160 | Small Engine | Actual=1.910 Pred=1.528

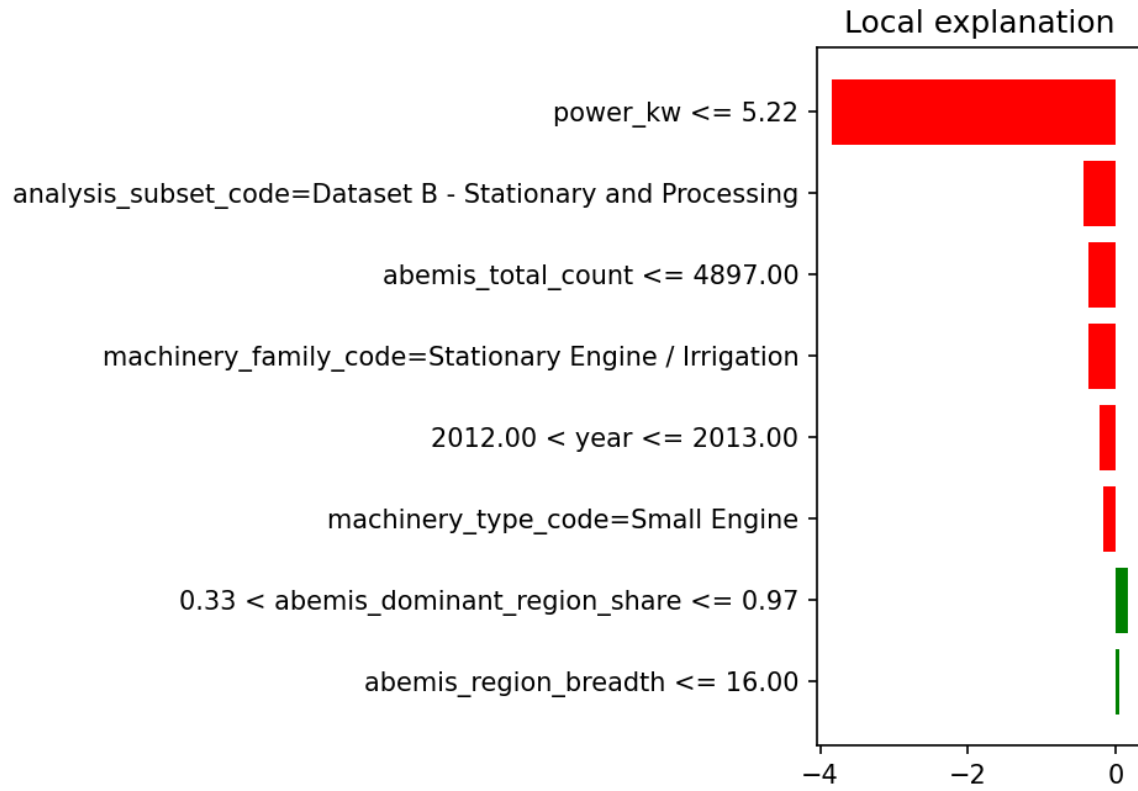


Fig. 23. LIME local explanation — example record 5.

LIME - Sample 260 | Combine Harvester | Actual=6.430 Pred=9.190

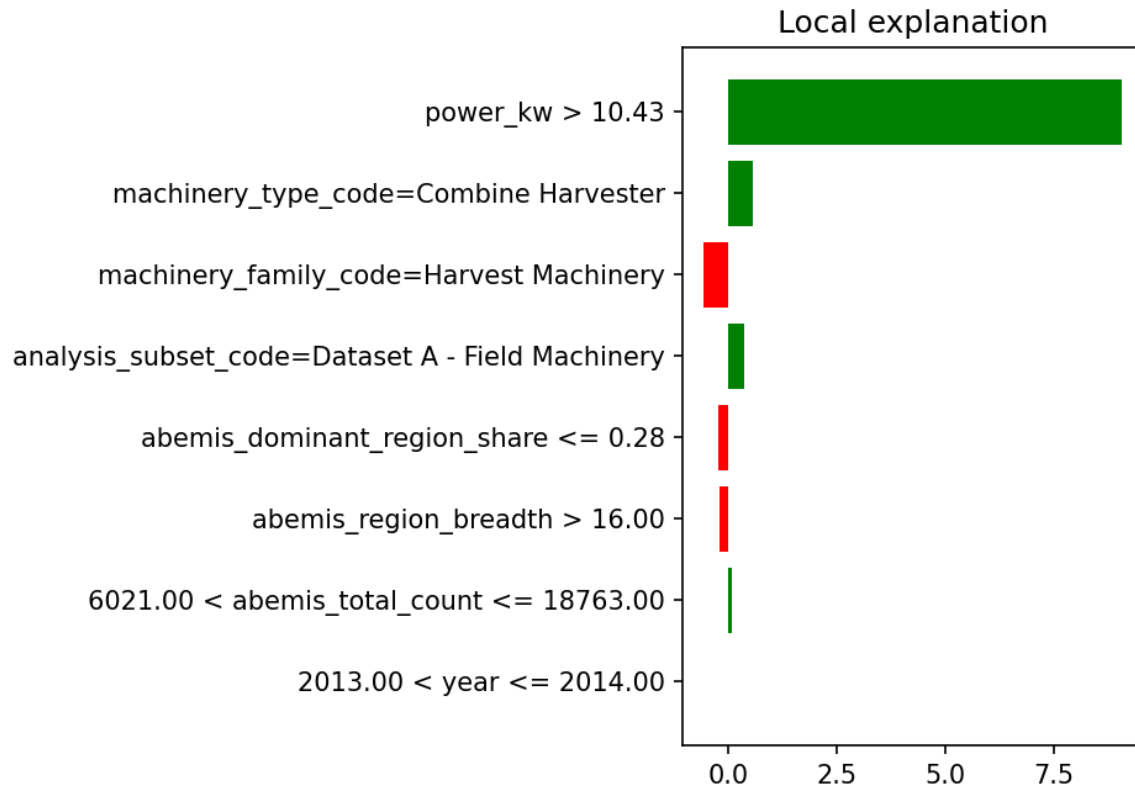


Fig. 24. LIME local explanation — example record 6.

LIME - Sample 35 | Thresher | Actual=1.900 Pred=1.716

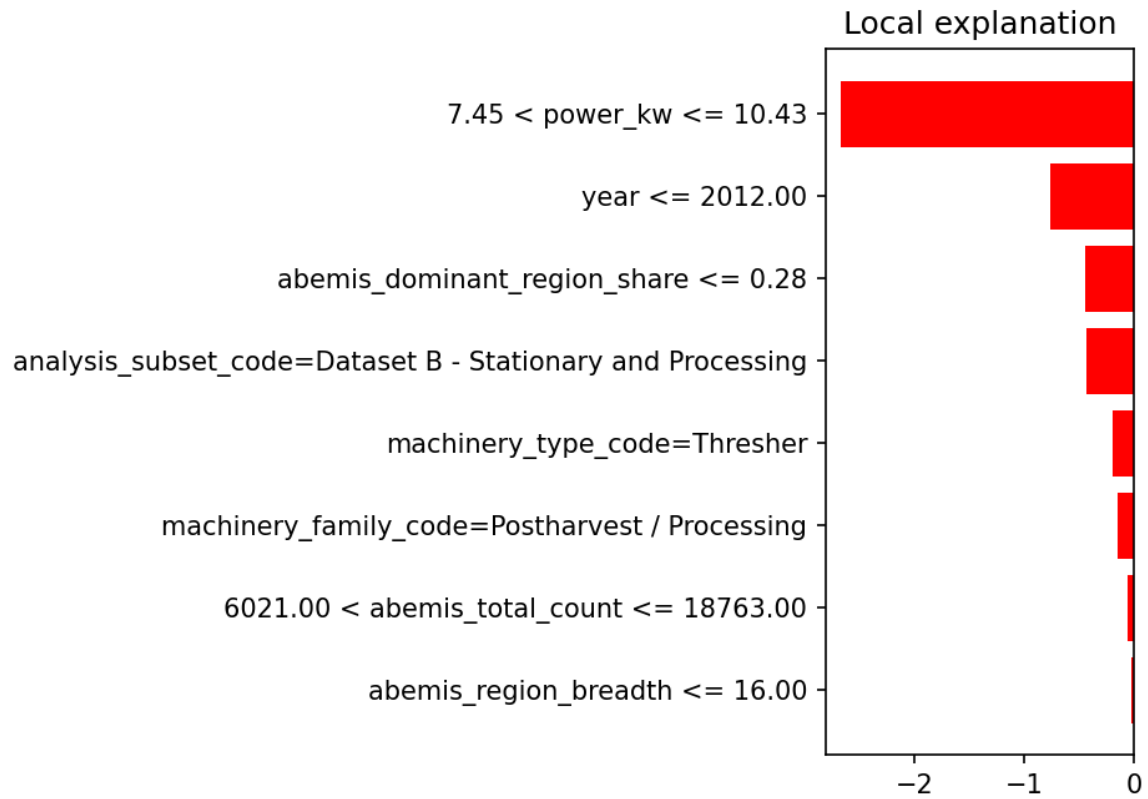


Fig. 25. LIME local explanation — example record 7.

LIME - Sample 75 | Thresher | Actual=4.650 Pred=3.889

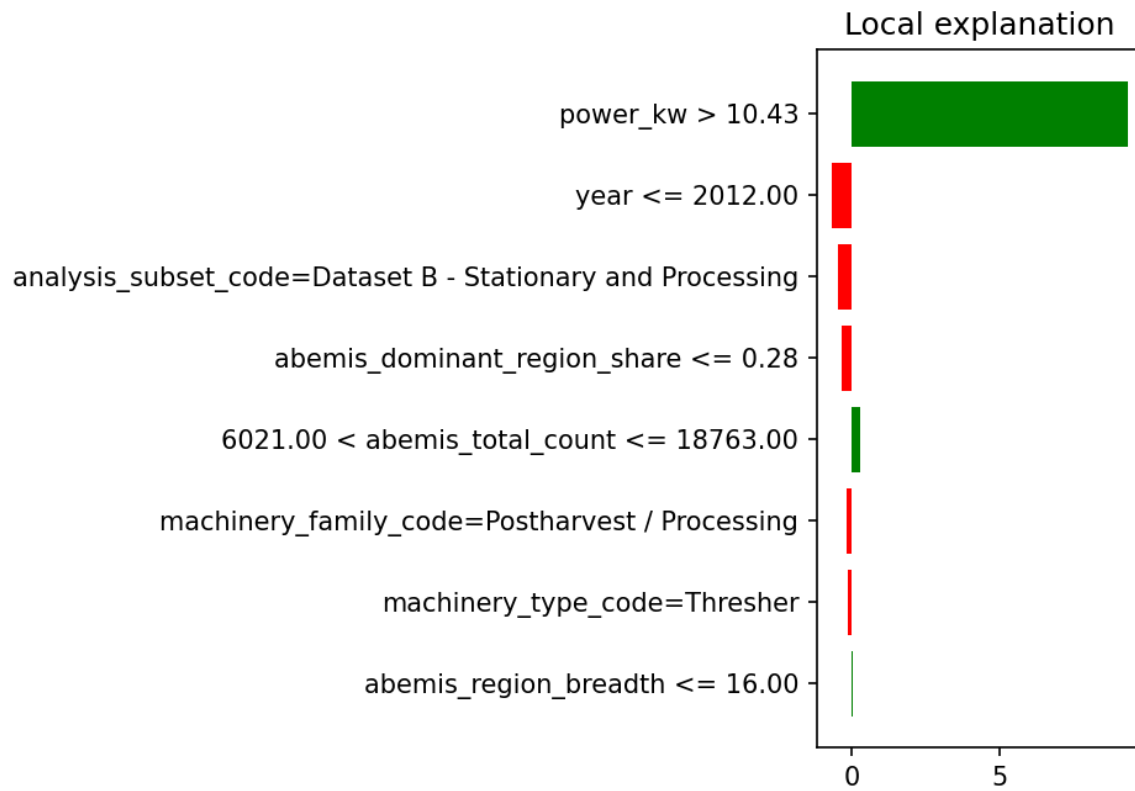


Fig. 26. LIME local explanation — example record 8.

LIME - Sample 319 | Four-Wheel Tractor | Actual=12.100 Pred=11.342

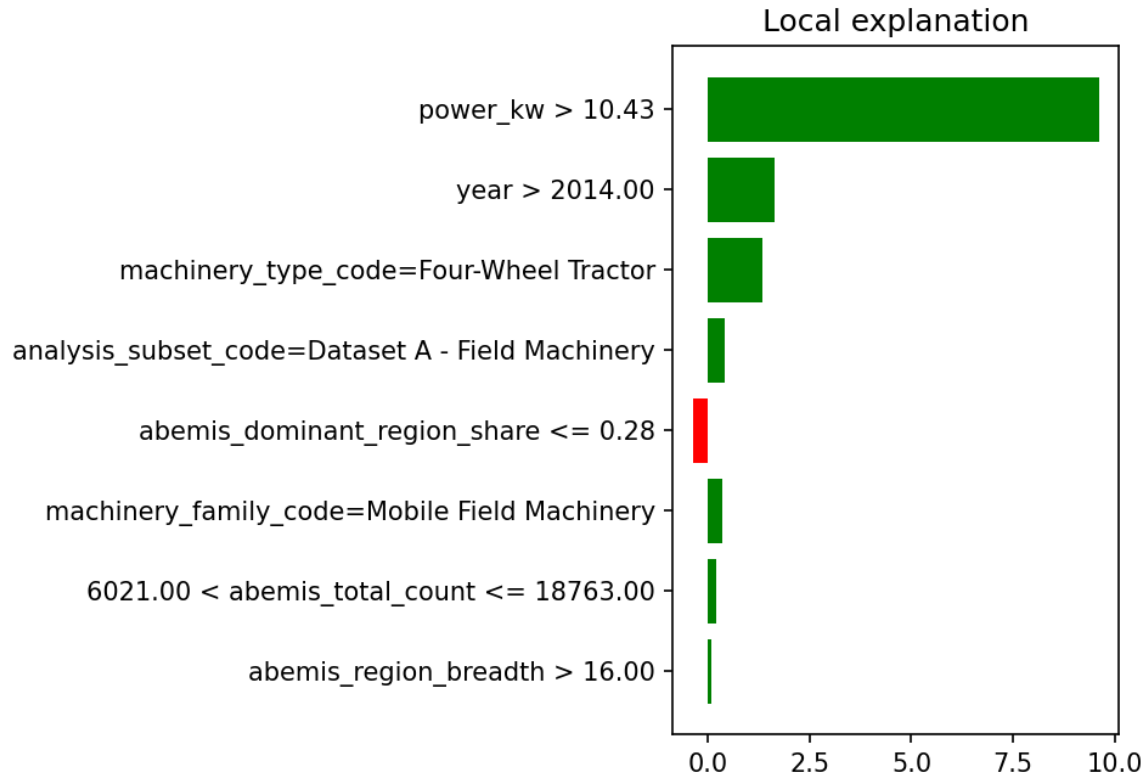


Fig. 27. LIME local explanation — example record 9.

Appendix D Descriptive Analytics

The following figures summarize the AMTEC dataset analytics used during exploratory data analysis and feature engineering. All figures are generated from the 376 cleaned AMTEC records.

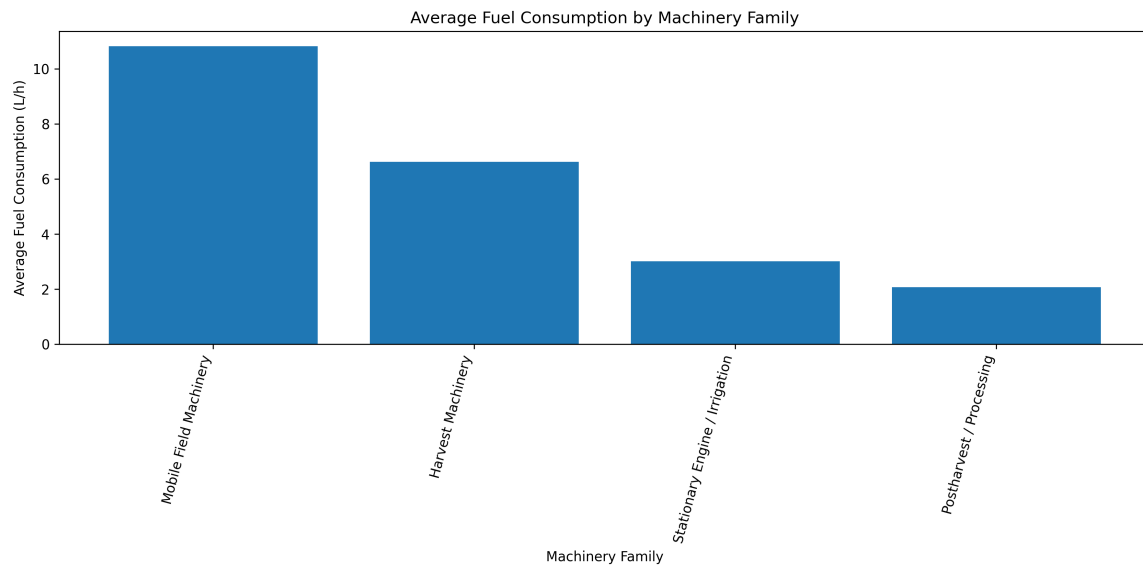


Fig. 28. Average fuel consumption (L/h) by machinery family. Harvest machinery and mobile field machinery show the highest average consumption, consistent with their typically larger engine power ratings.

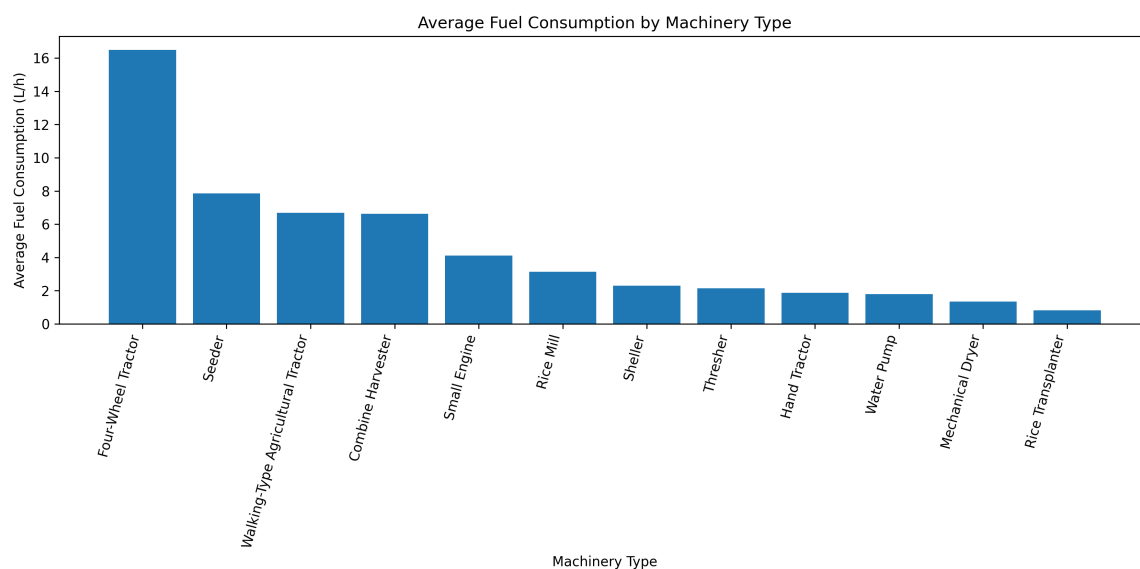


Fig. 29. Average fuel consumption (L/h) by machinery type. Combine harvesters and four-wheel tractors lead within their respective families.

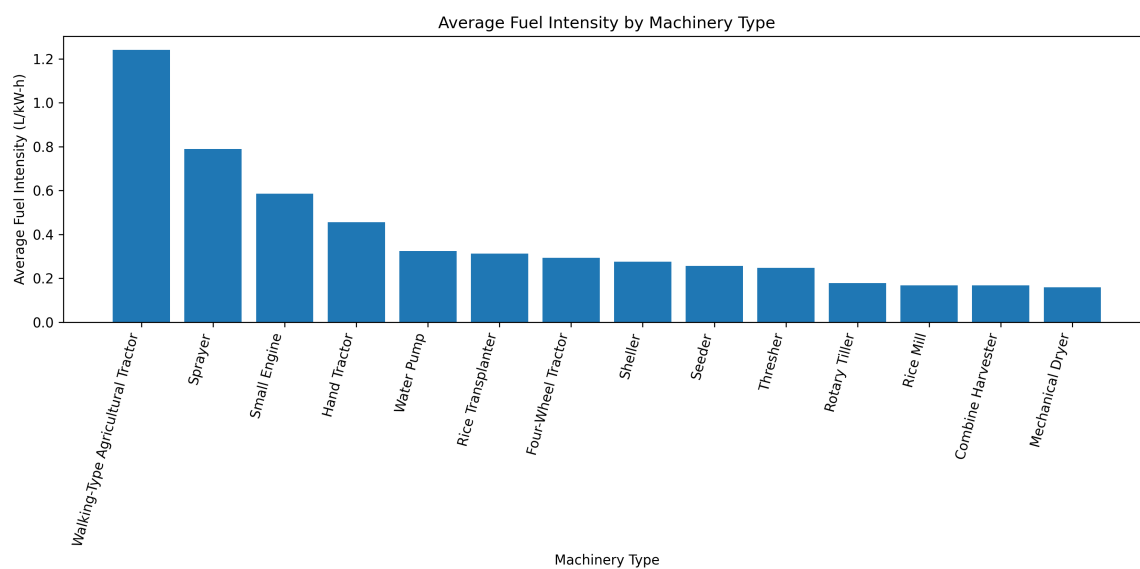


Fig. 30. Fuel intensity (L/h per kW rated power) by machinery type, highlighting operational efficiency differences across machinery categories.

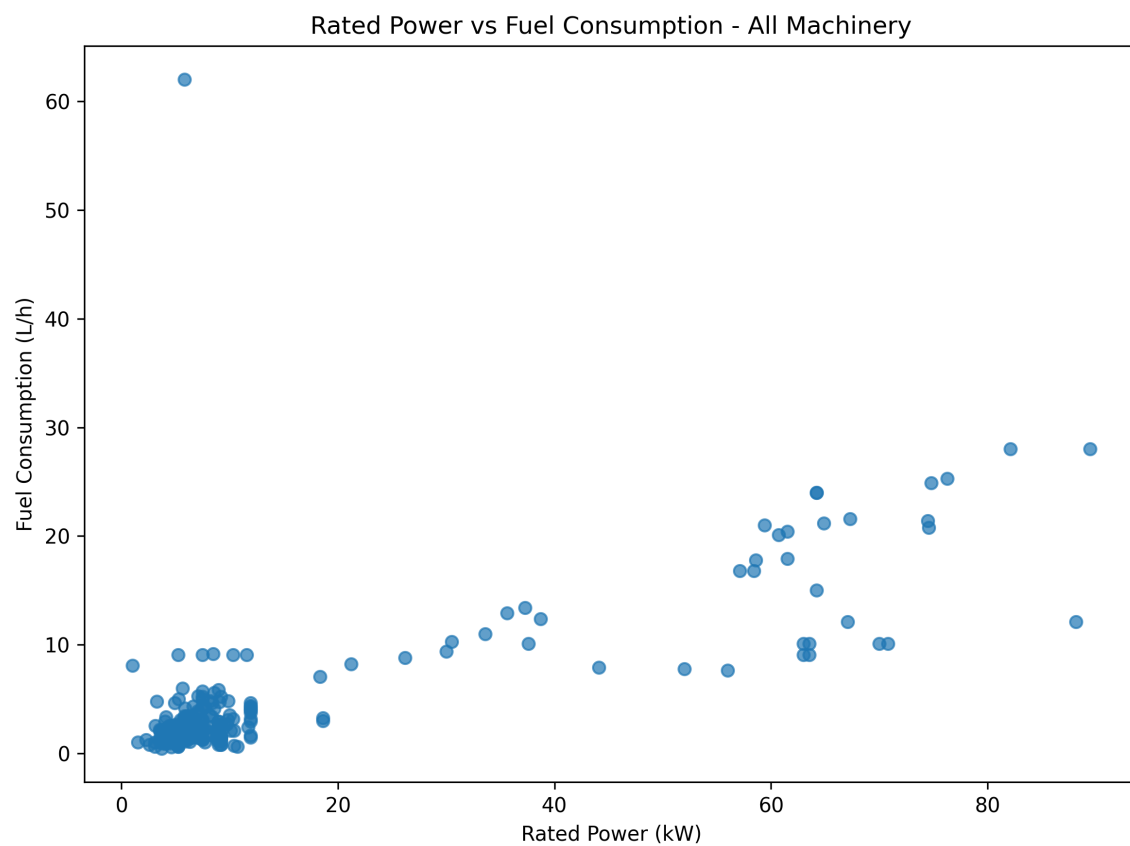


Fig. 31. Scatter plot of rated power (kW) vs. fuel consumption (L/h) for all 376 records. A strong positive relationship is visible, consistent with the SHAP finding that `power_kw` is the dominant predictor.

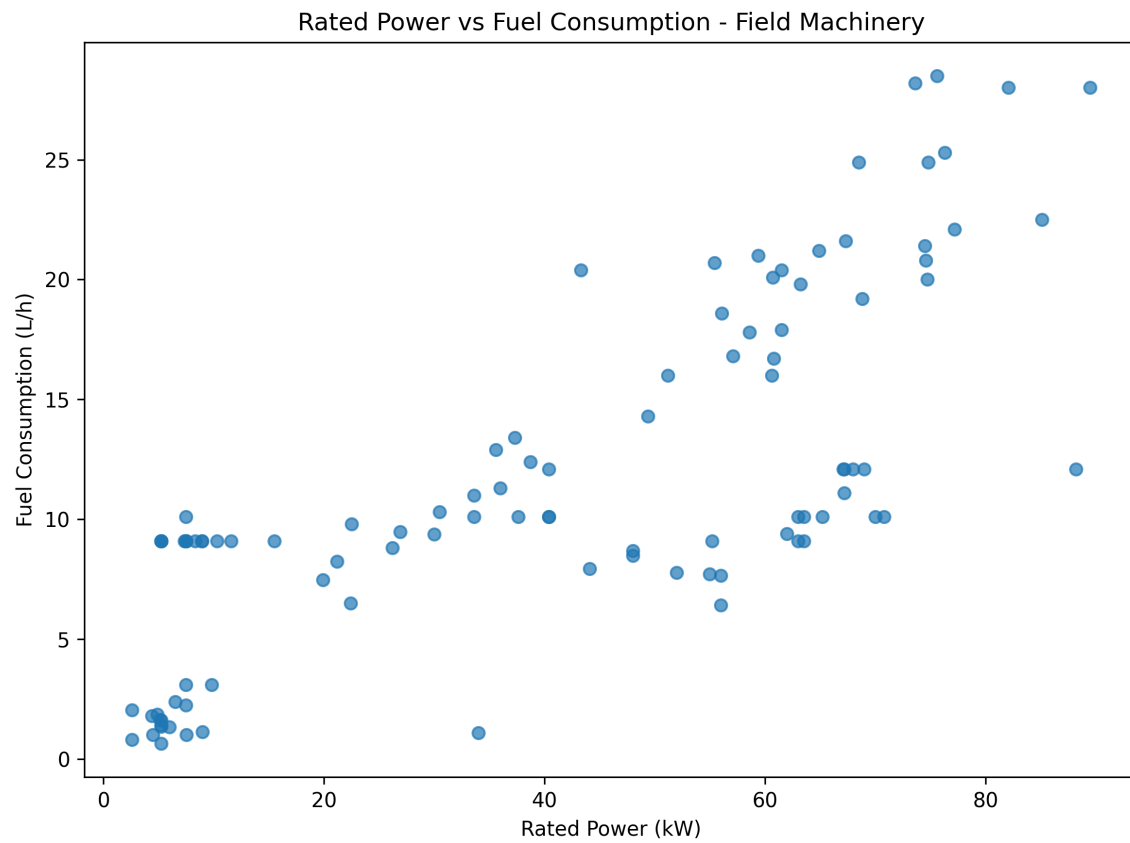


Fig. 32. Scatter plot of rated power (kW) vs. fuel consumption (L/h) for mobile field machinery only (tractors, tillers). The subset shows a tight linear trend, supporting the quadratic OLS enrichment within this group.